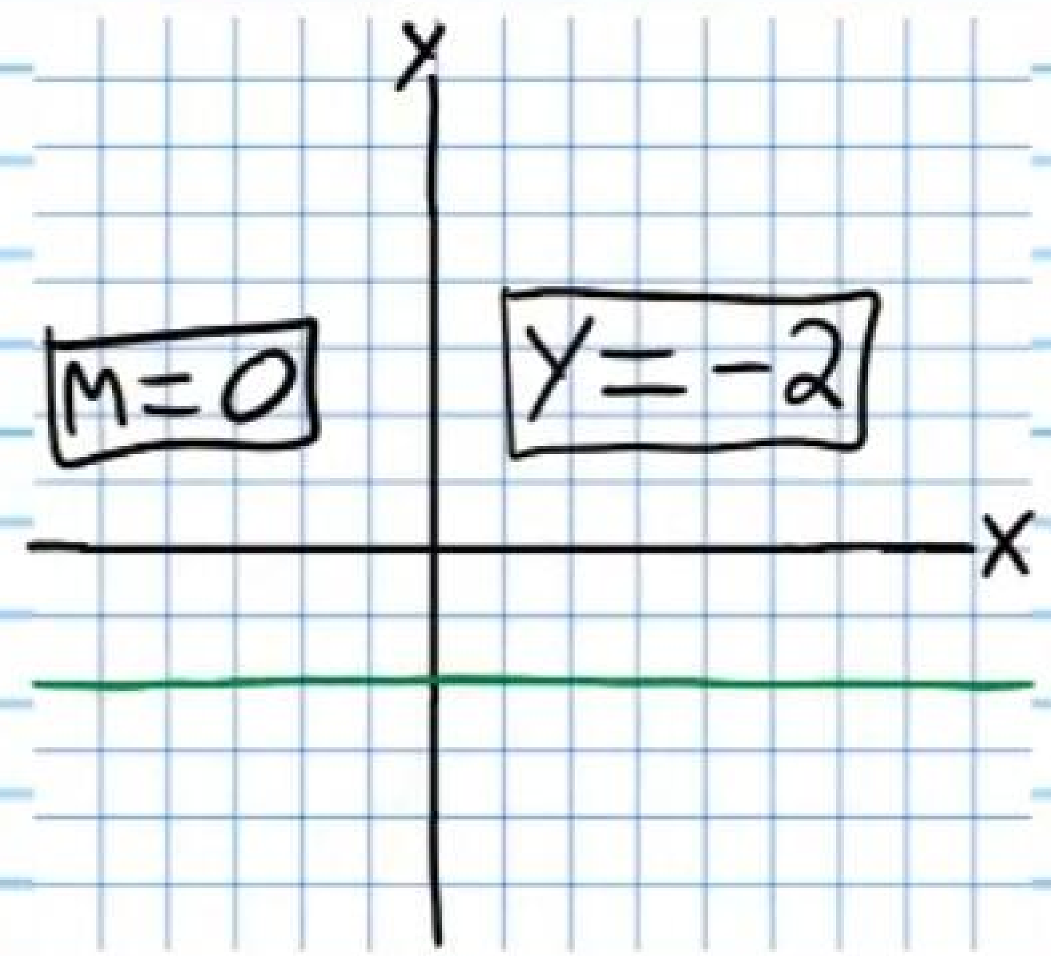
Name
Date
Course

Graphing Quadratic Expressions (Part III)

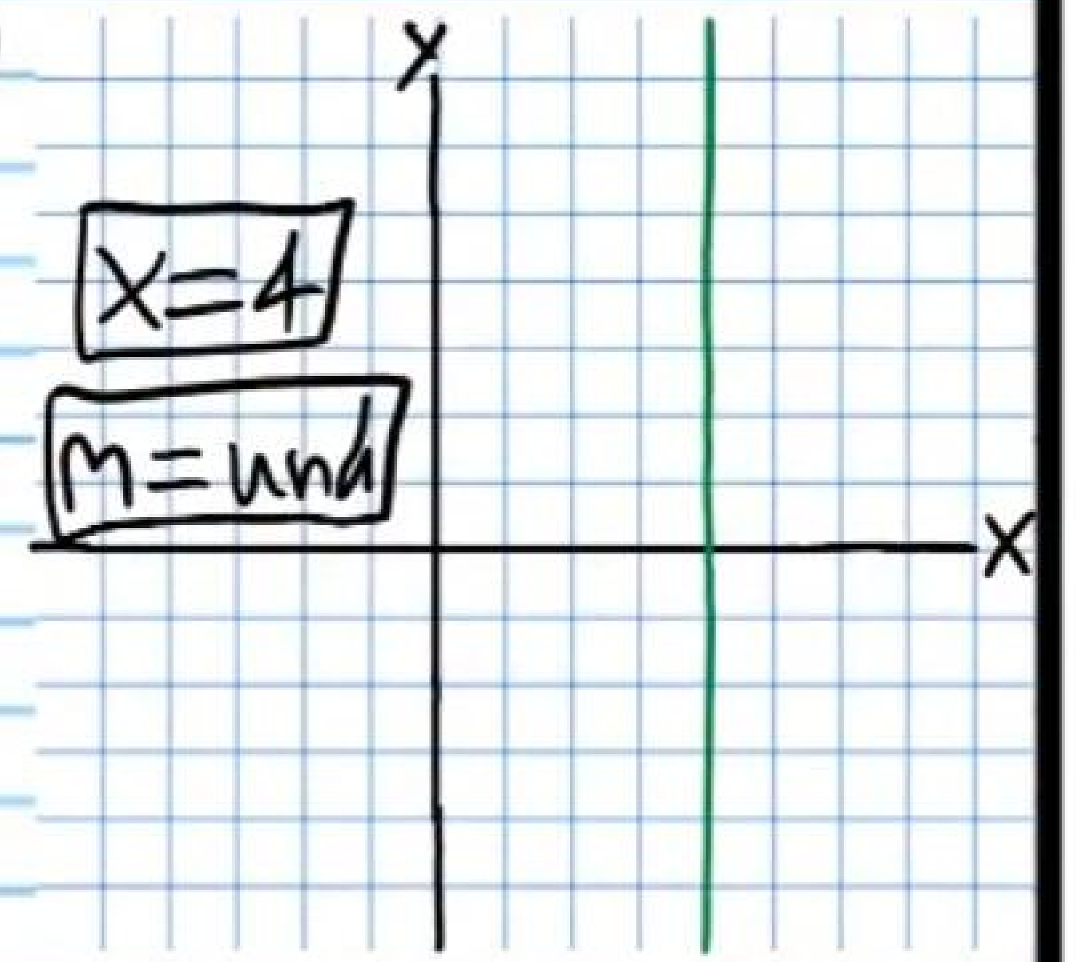
Review on Equations of Horizontal or Vertical Lines

Write equations of the lines graphed below and identify their slopes.

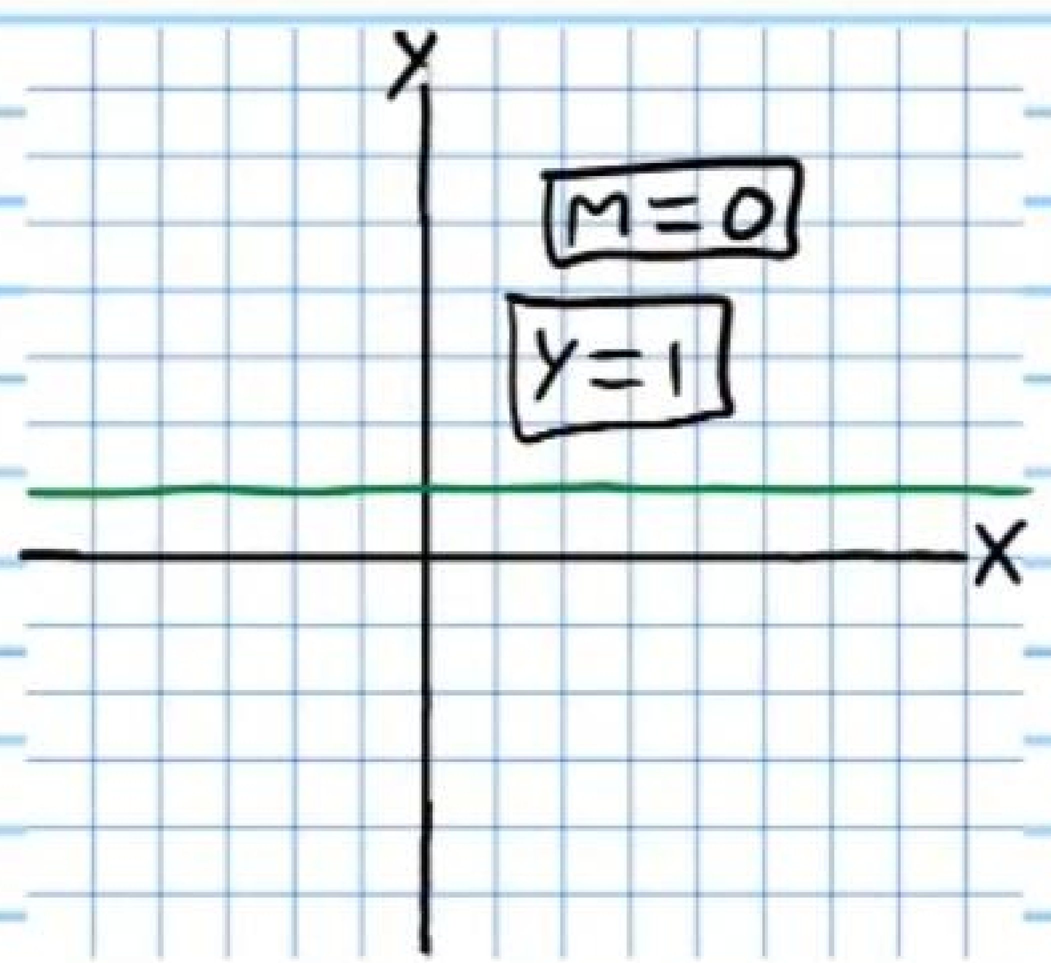
(1)



(2)



(3)



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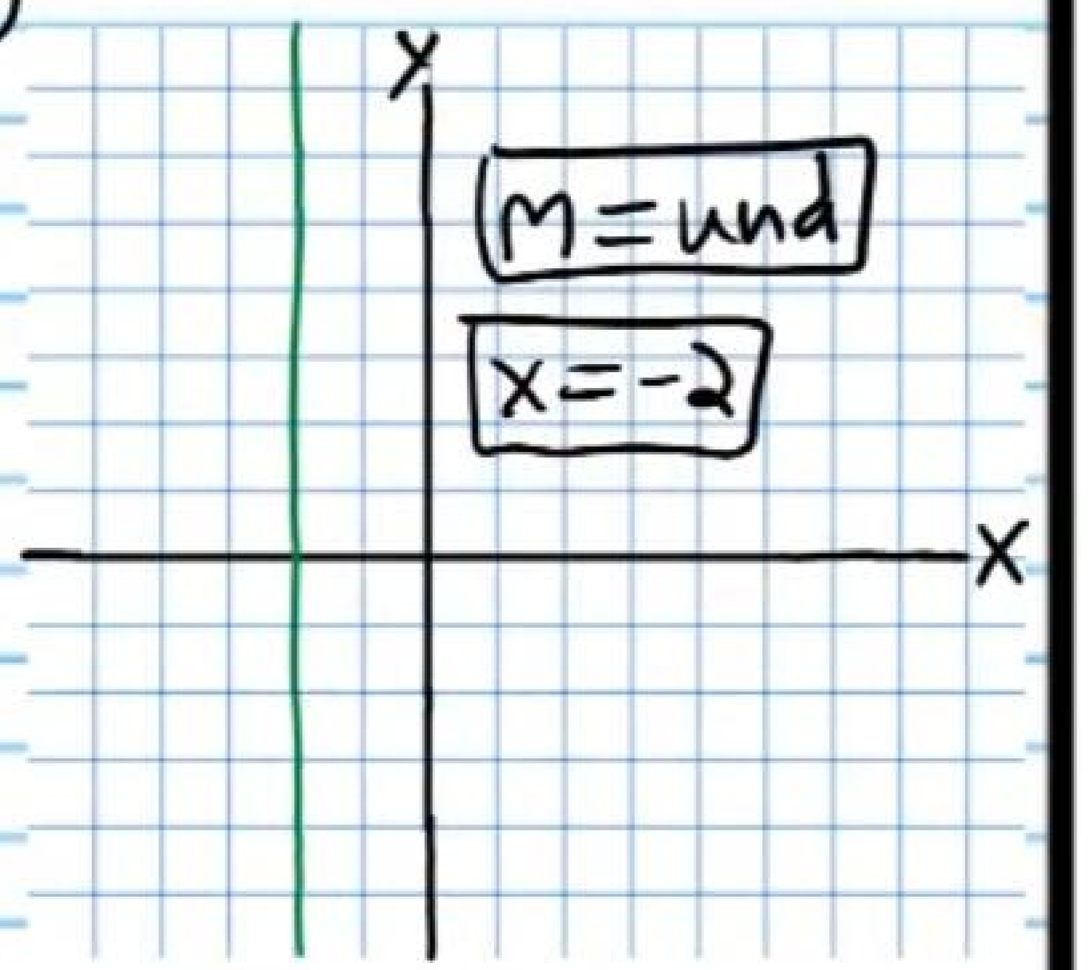
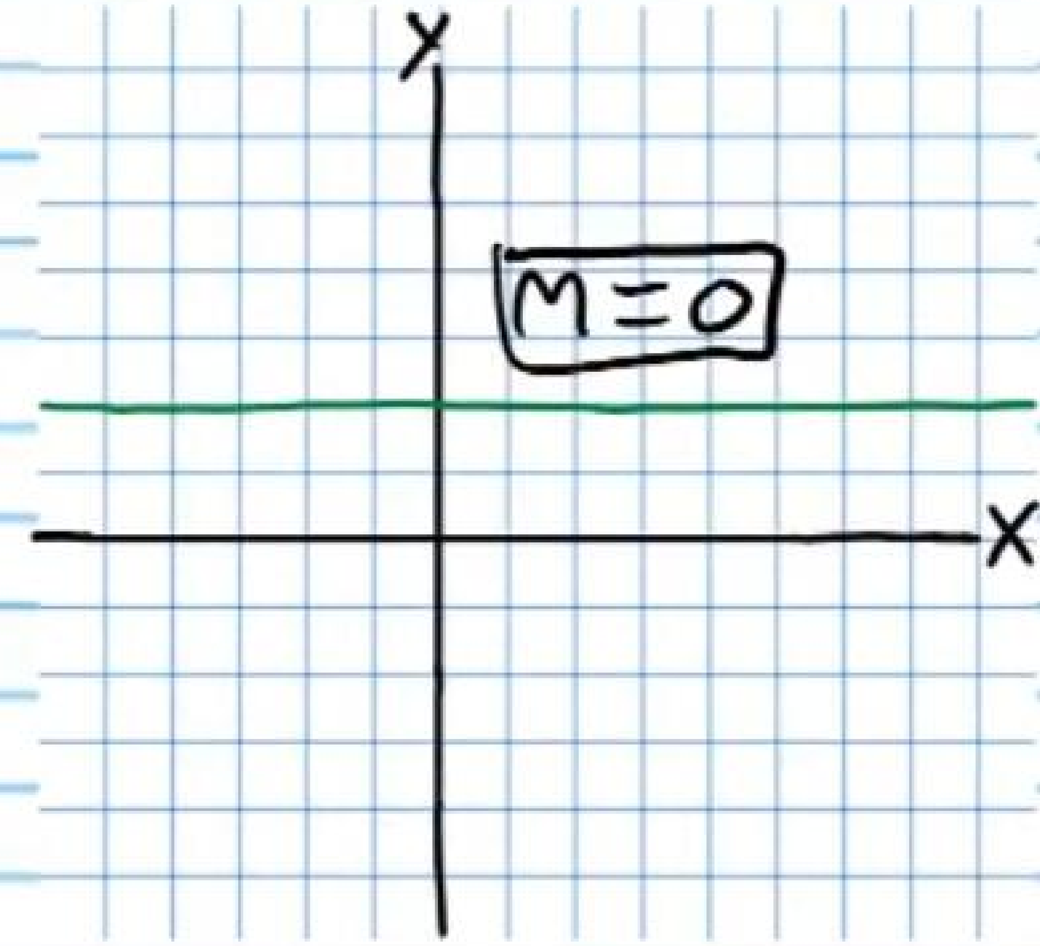
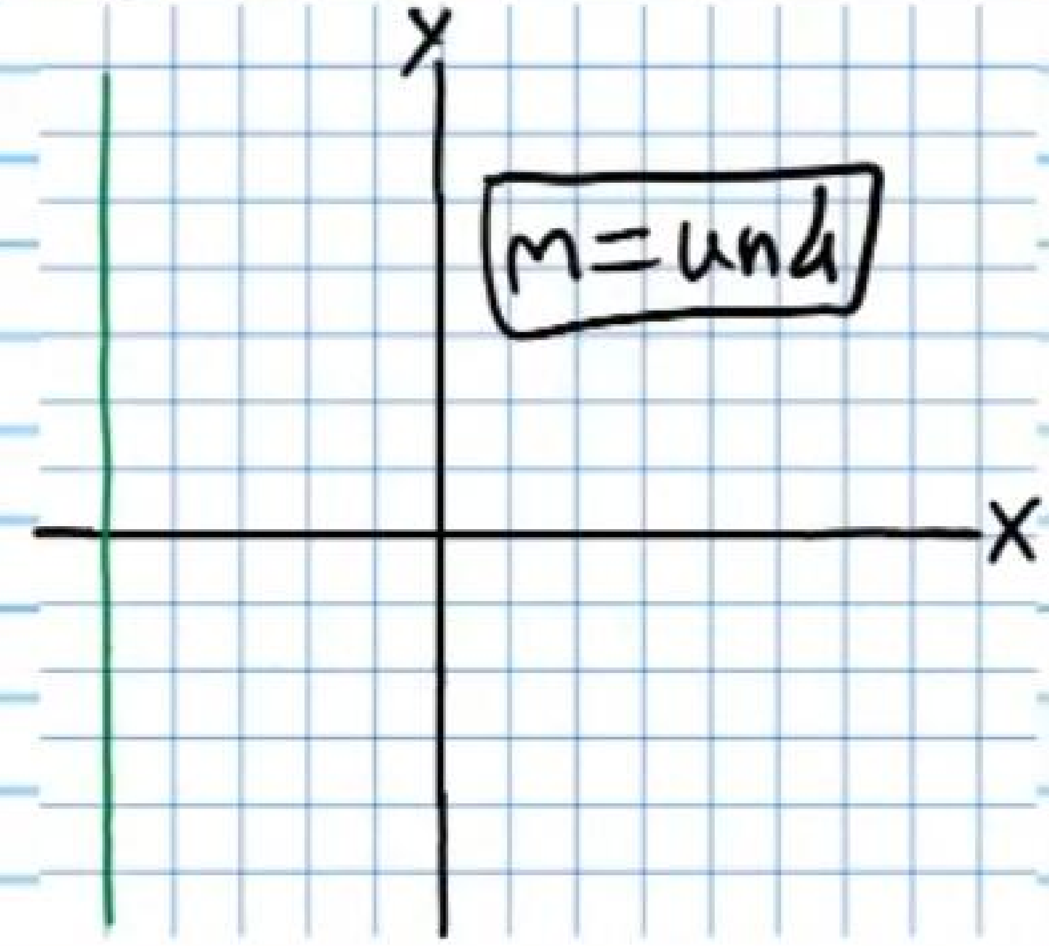


Graph the following equations and state the slopes of the resulting lines.

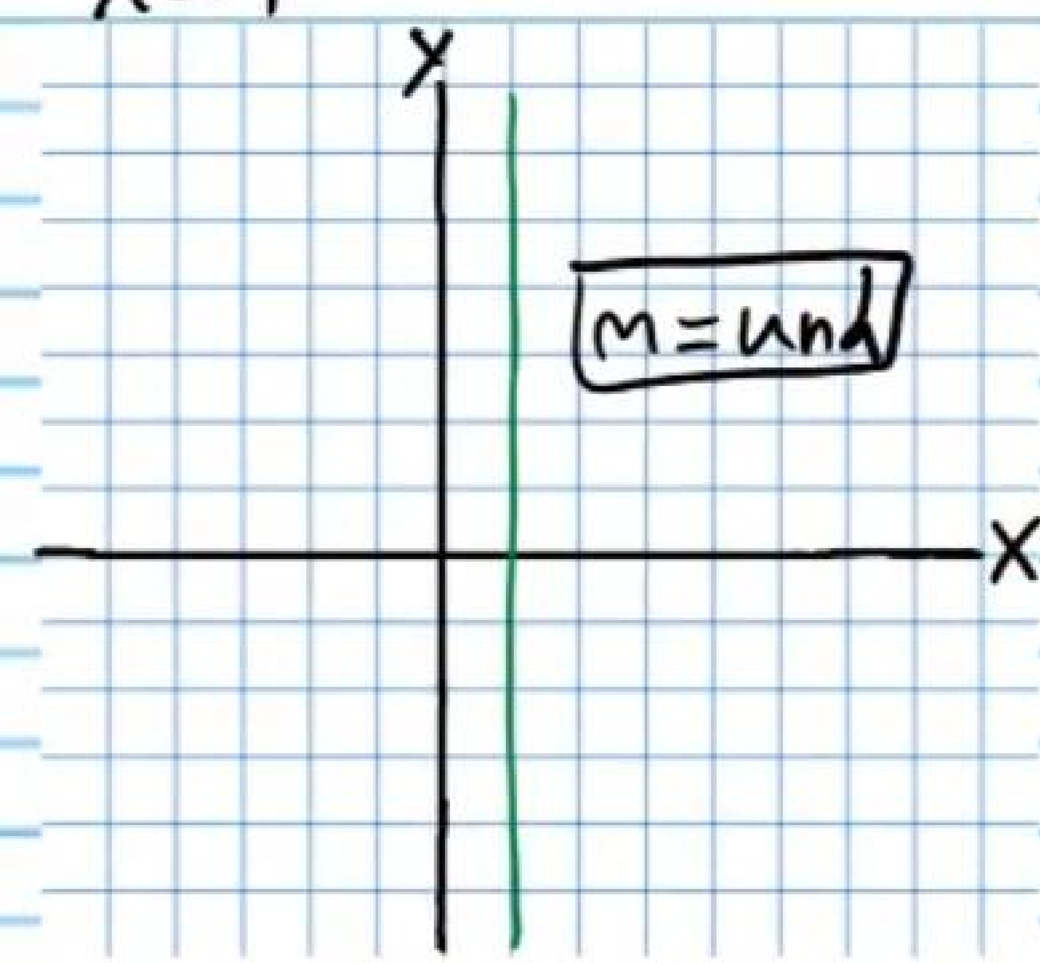
(5) $y=2$



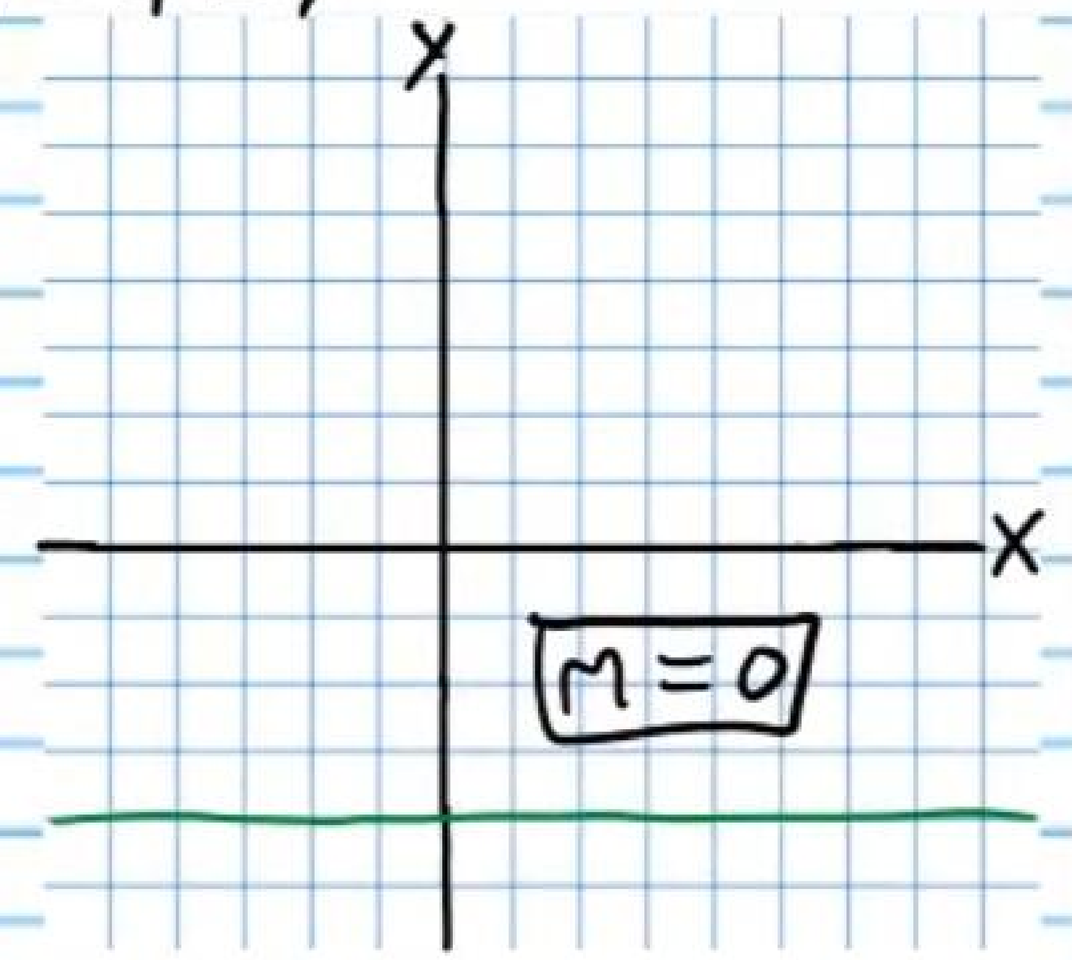
(6) $-5=x$



(7) $x=1$



(8) $-4=y$



Graphing Quadratic Expressions

Standard Form	Intercept Form	Vertex Form
$y = ax^2 + bx + c$	$y = a(x-m)(x-n)$	$y = a(x-h)^2 + k$

Finding the Vertex in Standard Form

If a quadratic expression is written in the form $y = ax^2 + bx + c$, then the x-value of the vertex is $-\frac{b}{2a}$.

Finding the Vertex in Intercept Form

If a quadratic expression is written in the form $y = a(x-m)(x-n)$, where m and n are real numbers, then the x-value of the vertex is $\frac{m+n}{2}$.

* Intercept form cannot be used to graph a quadratic expression if the corresponding graph does not move through the x-axis.

Finding the Vertex in Vertex Form

If a quadratic expression is written in the form $y = a(x-h)^2 + k$, then (h,k) is the vertex.

Graph each quadratic expression. Show the coordinates of the vertex, y-intercept, and x-intercepts (if there are any). Draw the line of symmetry (as a dotted line) and write its equation.

⑨ $y = 2(x+3)(x-1)$

$a = 2,\ m = -3,\ n = 1$

I) $V_x = \frac{m+n}{2} = \frac{-3+1}{2}$

$= -1$

$V_y = 2(-1+3)(-1-1)$

$= 2(2)(-2)$

$= -8$

$V(-1,-8)$

II) Y-Intercept

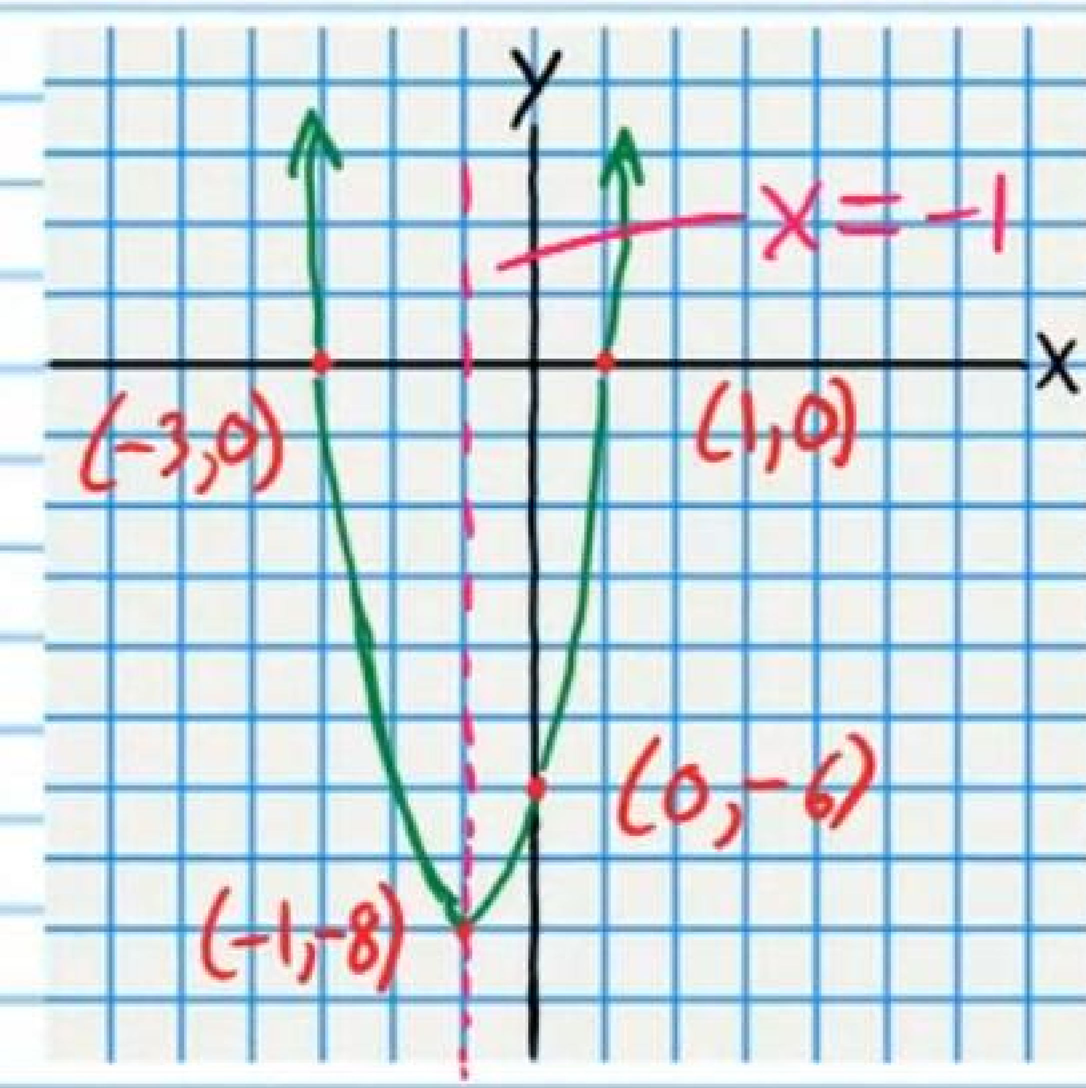
$x = 0$

$y = 2(0+3)(0-1)$

$= 2(3)(-1)$

$= -6$

$(0,-6)$



III) X-Intercepts

$(-3,0)$ & $(1,0)$

⑩ $-(x-2)(x-6)=y$

$a=-1,\ m=2,\ n=6$

I) $V_x = \frac{m+n}{2} = \frac{2+6}{2}$

$= 4$

$V_y = -(4-2)(4-6)$

$V_y = -(2)(-2)$

$= 4$

$V(4,4)$

II) Y-Intercept

$x = 0$

$y = -(0-2)(0-6)$

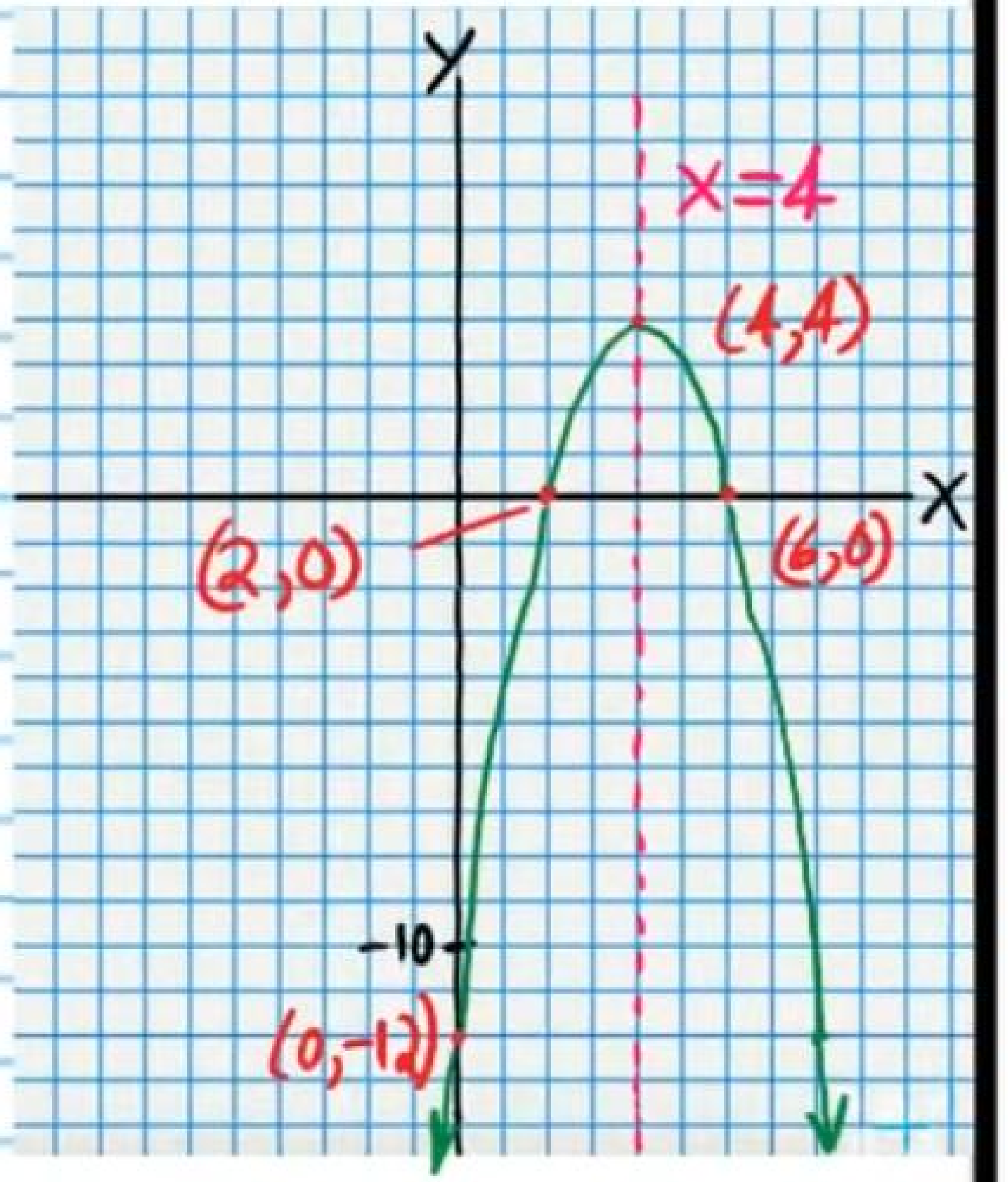
$= -(-2)(-6)$

$= -12$

$(0,-12)$

III) X-Intercepts

$(2,0)$ & $(6,0)$



⑪ $y=(x+1)(x+5)$

$a=1,\ m=-1,\ n=-5$

I) $V_x = \frac{m+n}{2} = \frac{-1+(-5)}{2}$

$= -3$

$V_y = (-3+1)(-3+5)$

$V_y = (-2)(2)$

$V_y = -4$

$V(-3,-4)$

II) Y-Intercept

$x = 0$

$y = (0+1)(0+5)$

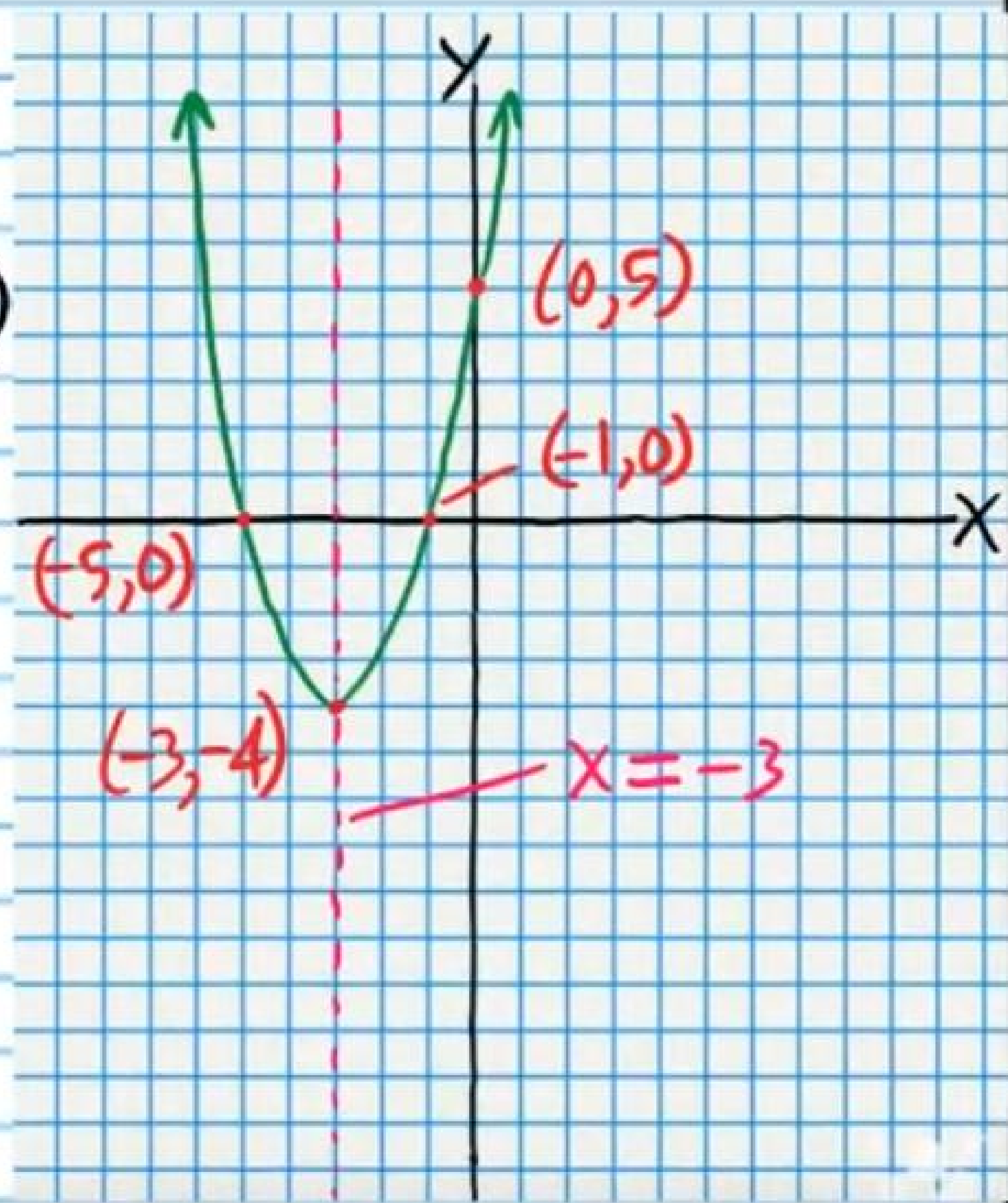
$y = (1)(5)$

$y = 5$

$(0,5)$

III) X-Intercepts

$(-1,0)$ & $(-5,0)$



⑫ $-2(x-2)(x-4)=y$

$a=-2,\ m=2,\ n=4$

I) $V_x = \frac{m+n}{2} = \frac{2+4}{2}$

$= 3$

$V_y = -2(3-2)(3-4)$

$V_y = -2(1)(-1)$

$V_y = 2$

$V(3,2)$

II) Y-Intercept

$x=0$

$y=-2(0-2)(0-4)$

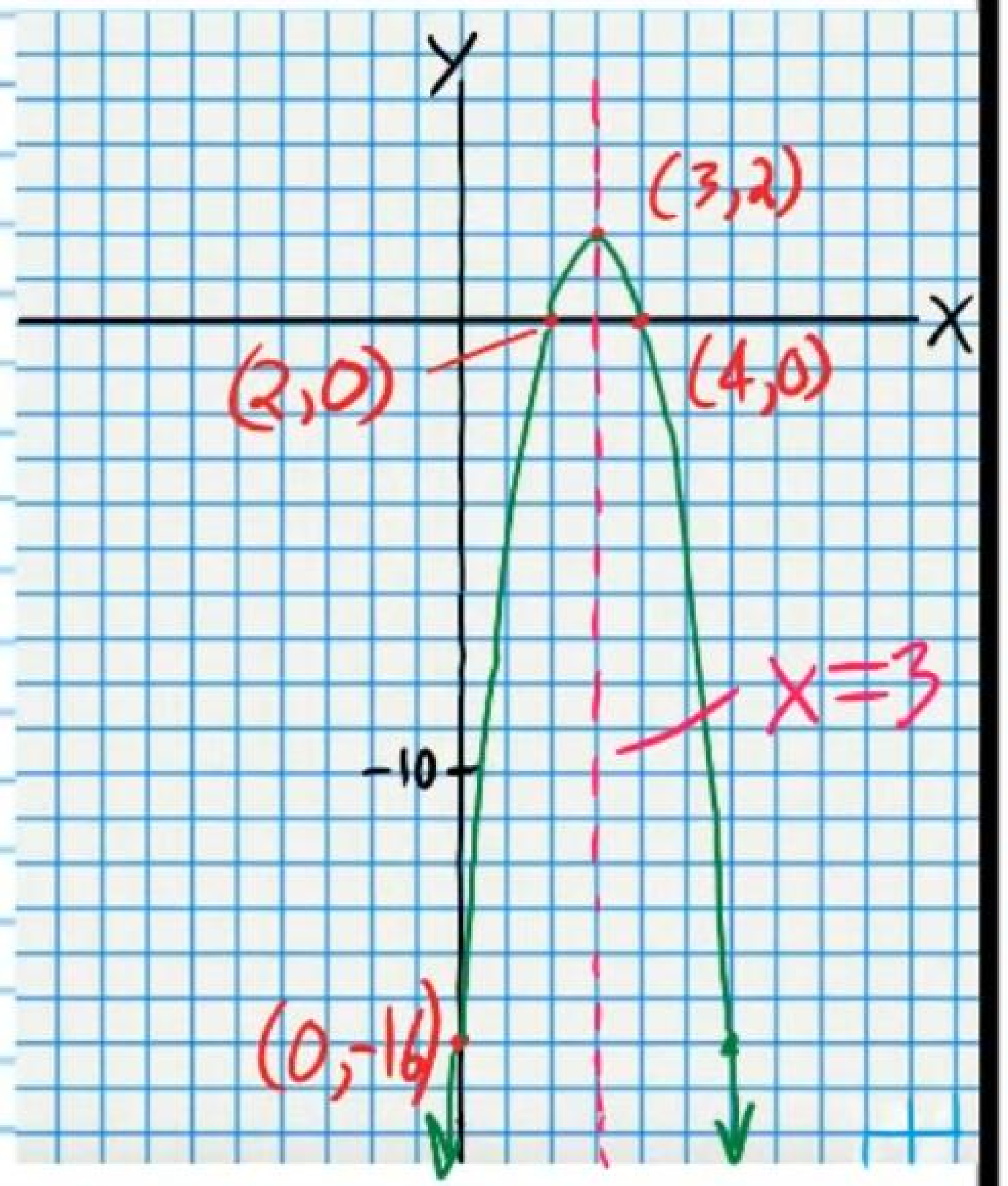
$y=-2(-2)(-4)$

$=-16$

$(0,-16)$

III) x-Intercepts

$(2,0)$ & $(4,0)$



⑬ $\frac{1}{3}(x+2)(x-2)=y$

$a=\frac{1}{3},\ m=-2,\ n=2$

I) $V_x = \frac{m+n}{2} = \frac{-2+2}{2}$

$= \frac{0}{2}$

$= 0$

$V_y = \frac{1}{3}(0+2)(0-2)$

$= \frac{1}{3}(2)(-2)$

$= \frac{1}{3}(-4)$

$= -\frac{4}{3}$

$= -1\frac{1}{3}$

$V(0,-1\frac{1}{3})$

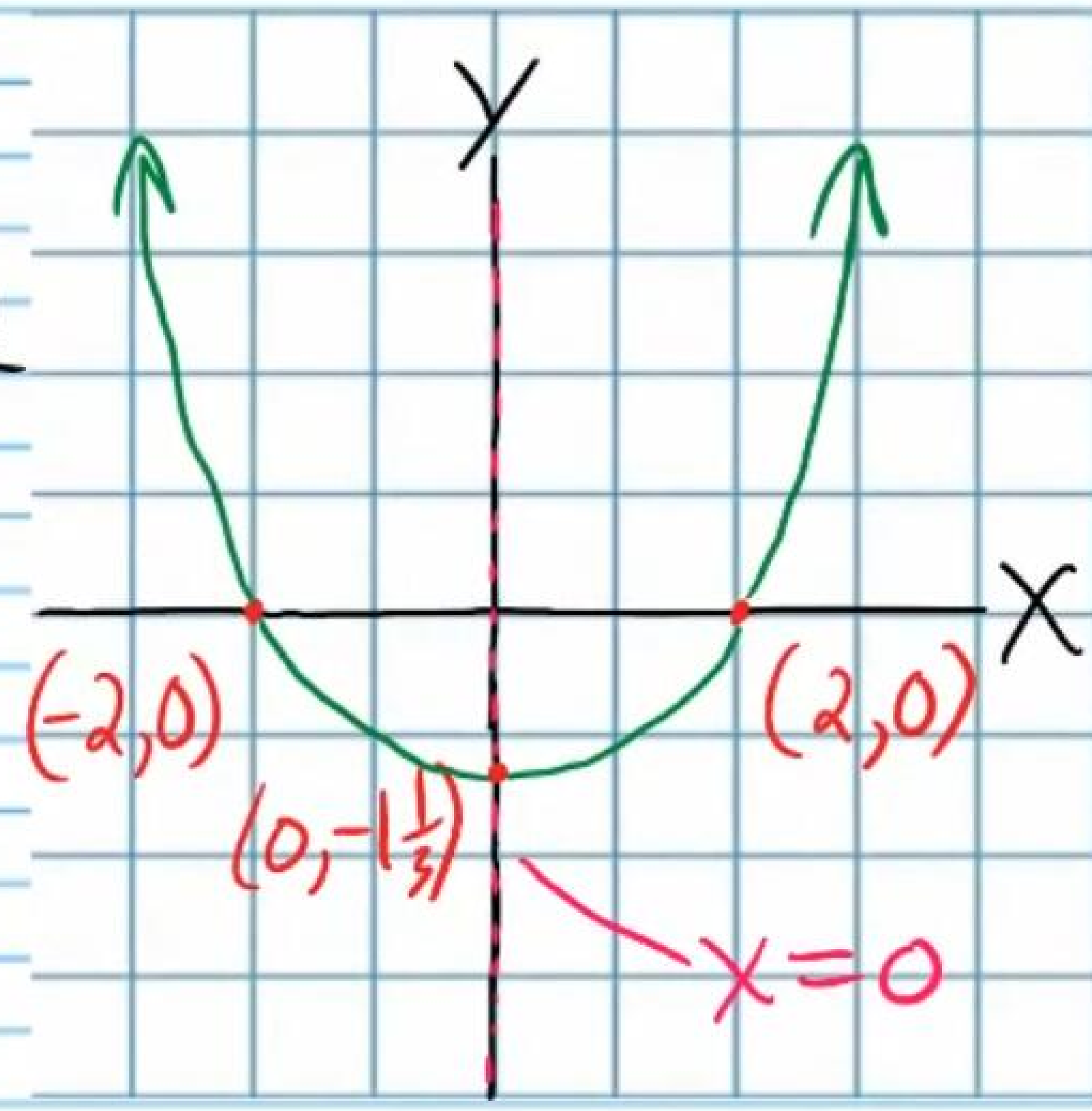
II) Y-Intercept

$x=0$

$(0,-1\frac{1}{3})$

III) X-Intercept

$(-2,0)$ & $(2,0)$



14) $y = -\frac{1}{2}(x-1)(x+1)$

$a = -\frac{1}{2}, m = 1, m = -1$

I) $V_x = \frac{m+n}{2} = \frac{1+(-1)}{2}$

$= \frac{0}{2}$

$= 0$

$V_y = -\frac{1}{2}(0-1)(0+1)$

$V_y = -\frac{1}{2}(-1)(1)$

$= \frac{1}{2}$

$V(0, \frac{1}{2})$

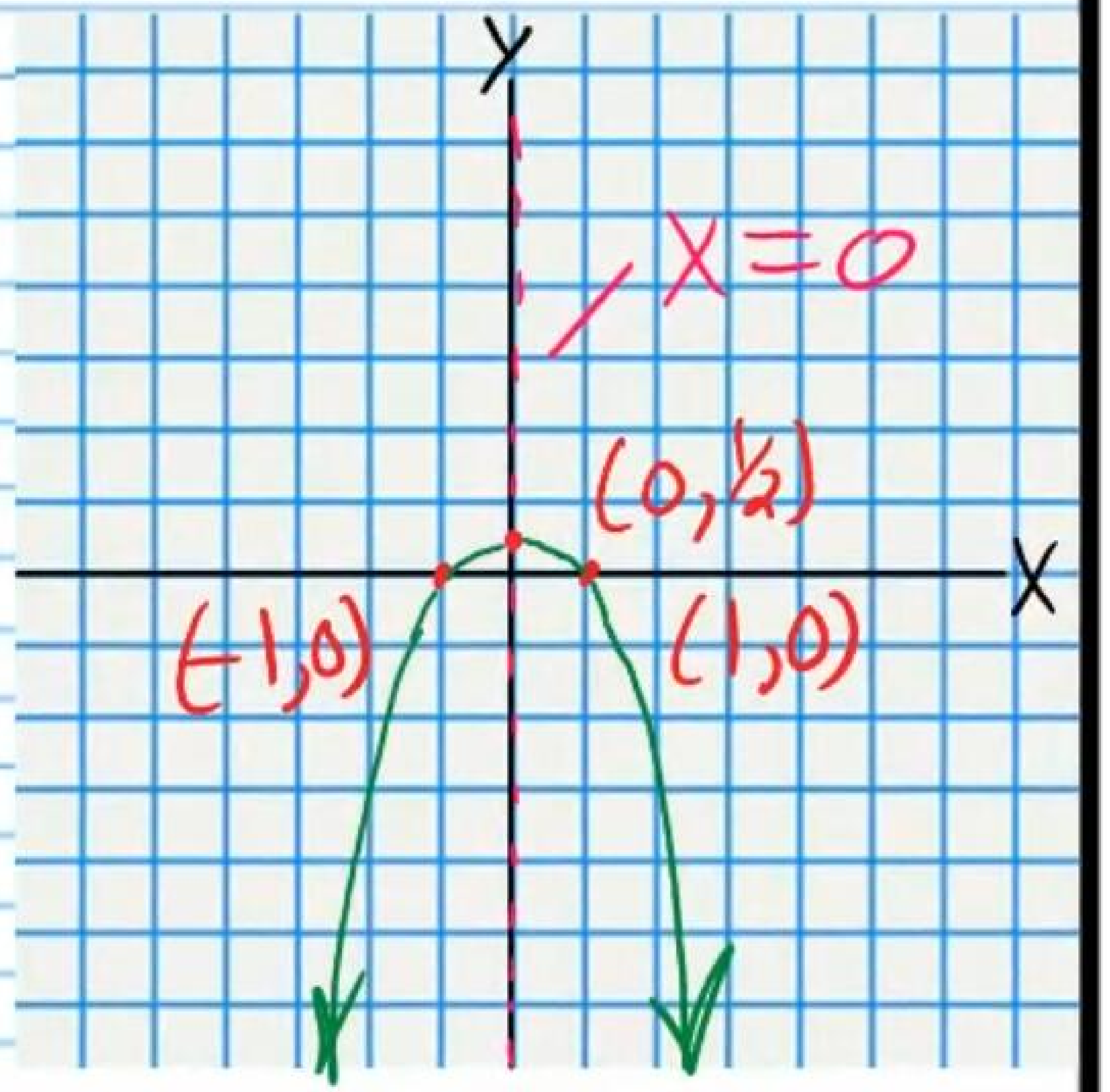
II) Y-Intercept

$x = 0$

$(0, \frac{1}{2})$

III) X-Intercepts

$(1,0)$ & $(-1,0)$



15) $y = -2(x-3)(x-3)$

$a = -2, m = 3, n = 3$

I) $V_x = \frac{m+n}{2} = \frac{3+3}{2}$

$= 3$

$V_y = -2(3-3)(3-3)$

$= -2(0)(0)$

$= 0$

$V(3,0)$

II) Y-Intercept

$x = 0$

$y = -2(0-3)(0-3)$

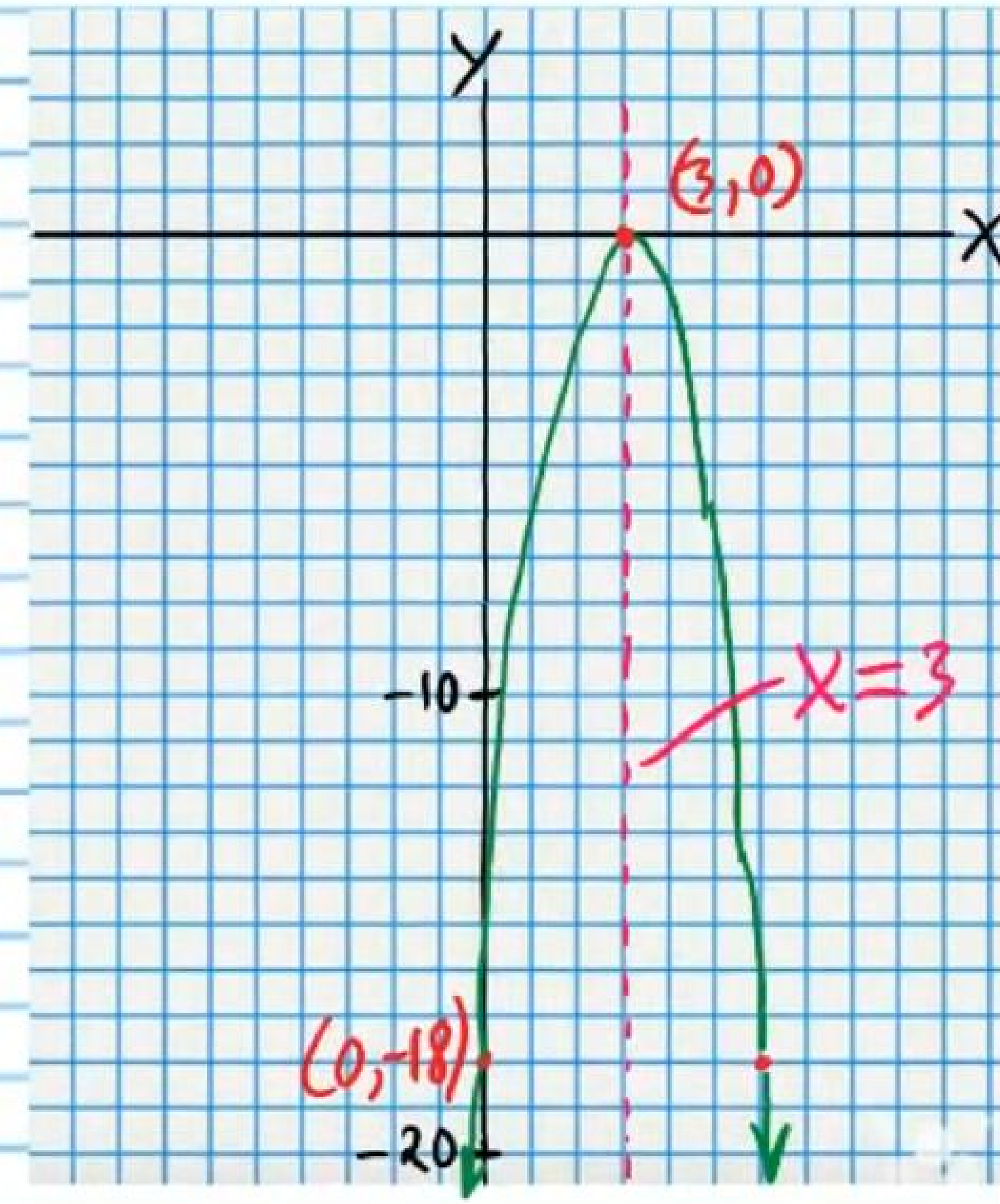
$y = -2(-3)(-3)$

$y = -18$

$(0,-18)$

III) X-Intercepts

$\underbrace{(3,0) \ \& \ (3,0)}_{\text{The same}}$



⑯ $3(x+2)(x+2)=y$

$a=3,\ m=-2,\ n=-2$

I) $V_x = \frac{m+n}{2} = \frac{-2+(-2)}{2}$

$= -2$

$V_y = 3(-2+2)(-2+2)$

$= 3(0)(0)$

$= 0$

$V(-2,0)$

II) Y-Intercept

$x = 0$

$y = 3(0+2)(0+2)$

$y = 3(2)(2)$

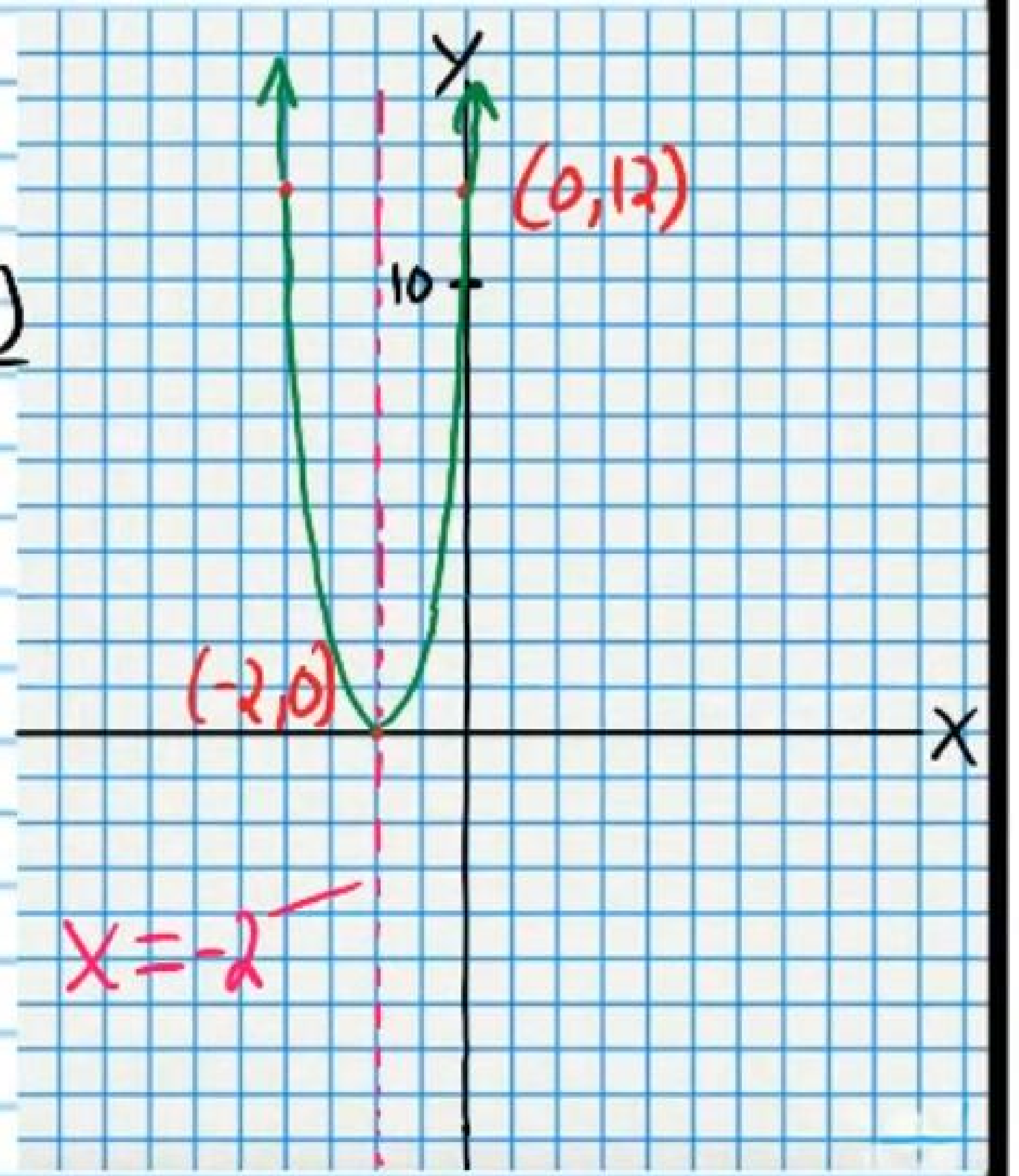
$y = 12$

$(0,12)$

III) X-Intercepts

$(-2,0)$ & $(-2,0)$

The same



⑰ $y = 3x^2 + 12x$

$y = 3x(x+4)$

$y = 3(x-0)(x+4)$

$a=3,\ m=0,\ n=-4$

I) $V_x = \frac{m+n}{2} = \frac{0+(-4)}{2}$

$= -2$

$V_y = 3(-2)(-2+4)$

$V_y = 3(-2)(2)$

$V_y = -12$

$V(-2,-12)$

II) Y-Intercept

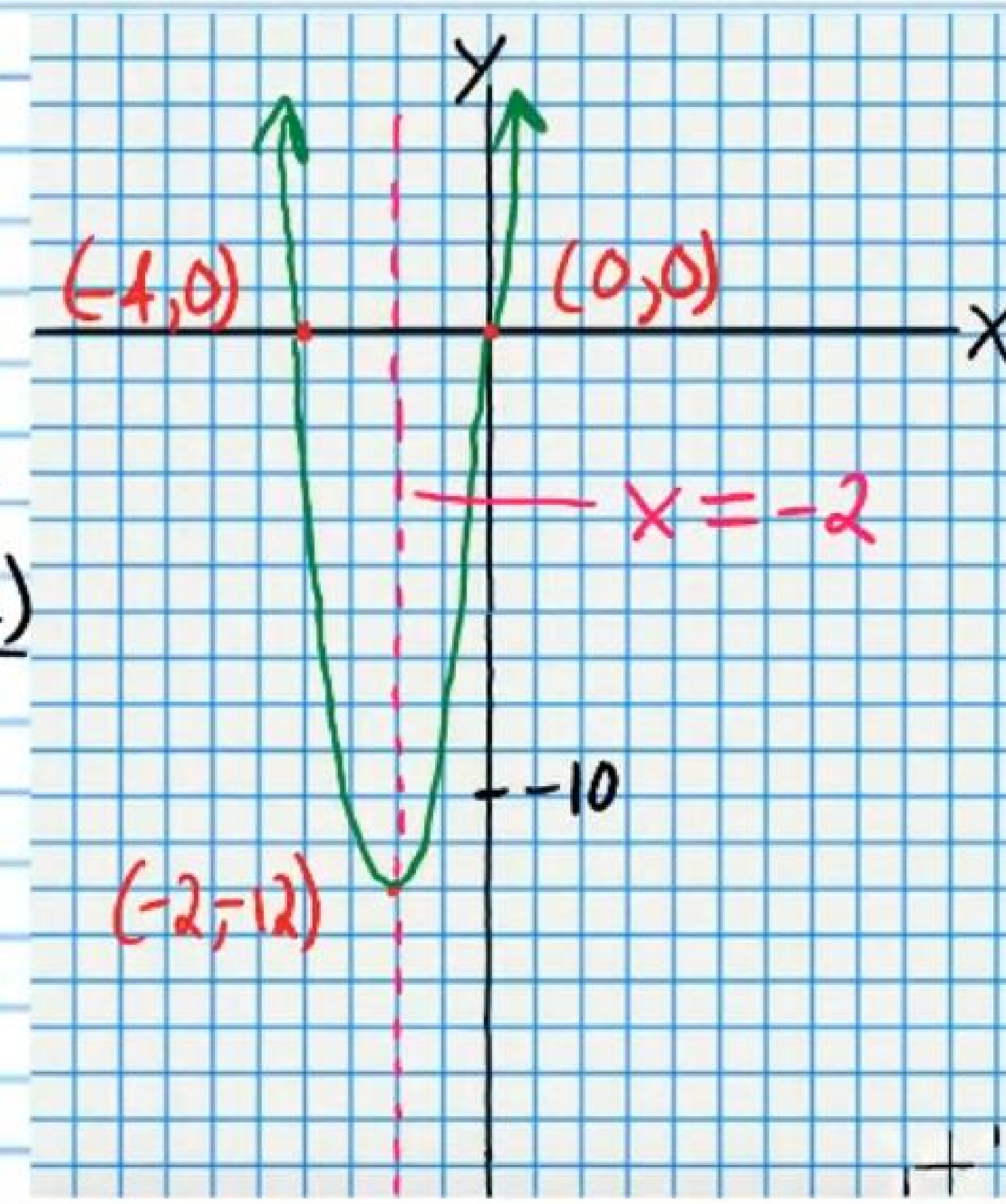
$x = 0$

$y = 3(0)(0+4)$

$= 3(0)(4)$

$= 0$

$(0,0)$



III) X-Intercepts

$(0,0)$ & $(-4,0)$

(18) $x^2-6x=y$

$x(x-6)=y$

$(x-0)(x-6)=y$

$a=1,\ m=0,\ n=6$

I) $V_x=\frac{m+n}{2}=\frac{0+6}{2}$

$=3$

$V_y=3(3-6)$

$V_y=3(-3)$

$V_y=-9$

$V(3,-9)$

II) Y-Intercept

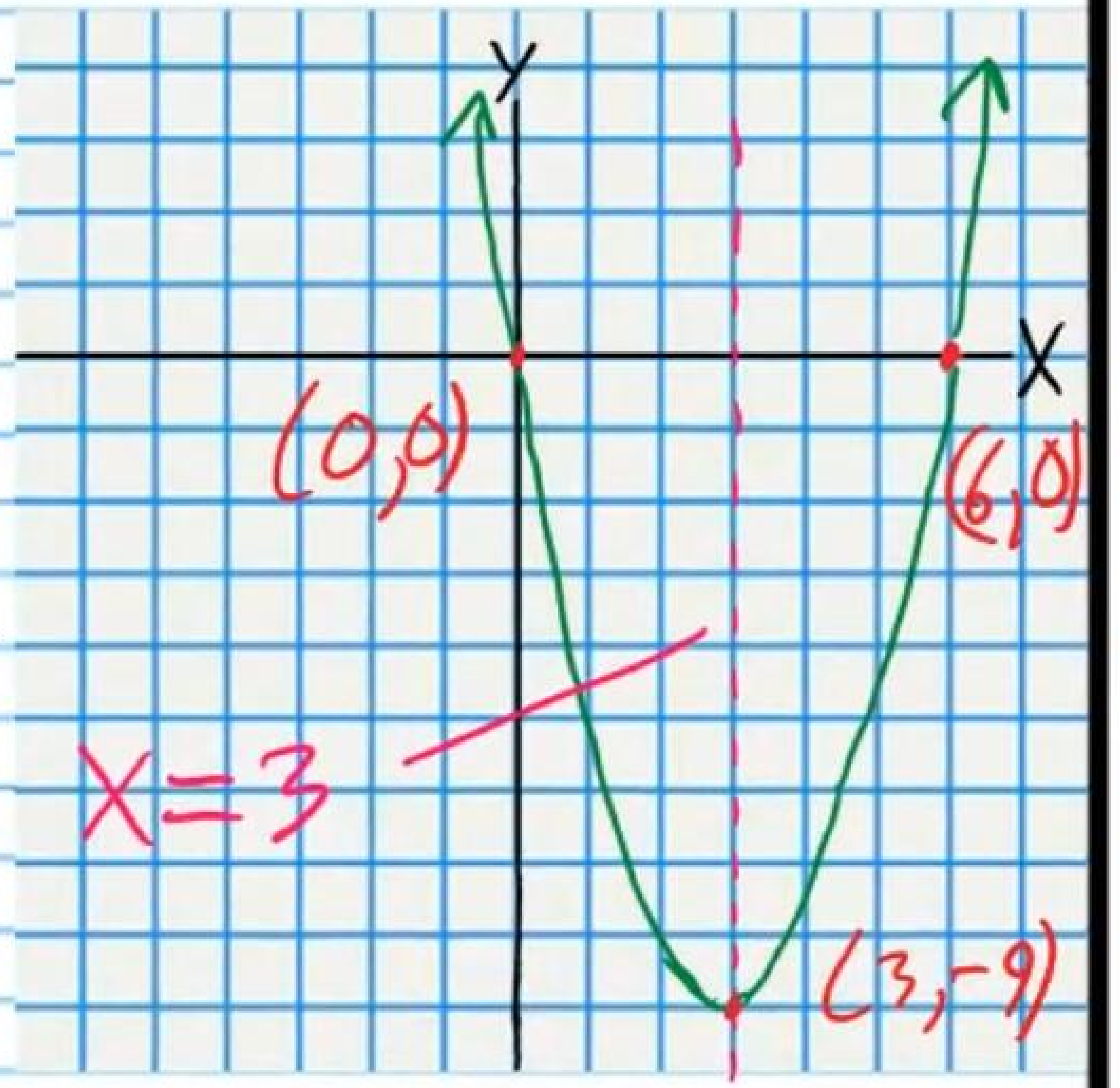
$x=0$

$y=0(0-6)$

$y=0(-6)$

$y=0$

$(0,0)$



III) X-Intercepts

$(0,0)$ & $(6,0)$

(19) $4(x+2)^2-4=y$

$a=4,\ h=-2,\ k=-4$

I) $V_x=h=-2,\ V_y=k=-4$

$V(-2,-4)$

II) Y-Intercept

$x=0$

$y=4(0+2)^2-4$

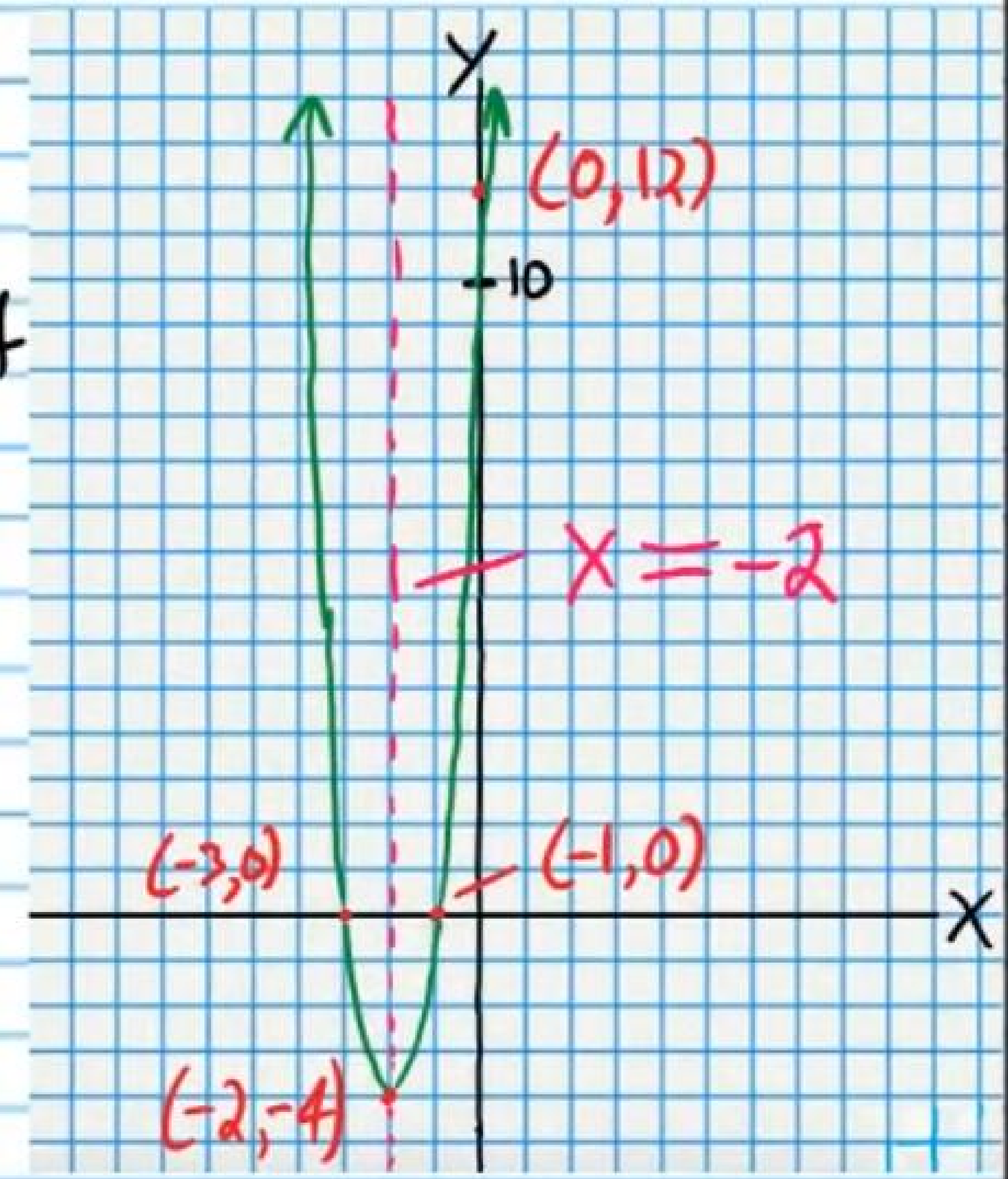
$y=4(2)^2-4$

$y=4(4)-4$

$y=16-4$

$y=12$

$(0,12)$



III) X-Intercepts

$y=0$

$4(x+2)^2-4=0$

$4(x+2)^2=4$

$(x+2)^2=1$

$\sqrt{(x+2)^2}=\pm\sqrt{1}$

$x+2=\pm 1$

$x=-2\pm 1$

$x=-2+1$ or $x=-2-1$

$x=-1$ or $x=-3$

$(-1,0)$ & $(-3,0)$

⑳ $\frac{1}{3}(x+5)^2=y$

$\frac{1}{3}(x+5)^2+0=y$

$a=\frac{1}{3},\ h=-5,\ k=0$

I) $V_x=h=-5$

$V_y=k=0$

$V(-5,0)$

II) Y-Intercept

$x=0$

$y=\frac{1}{3}(0+5)^2$

$y=\frac{1}{3}(5)^2$

$y=\frac{1}{3}(25)$

$y=\frac{25}{3}$

$y=8\frac{1}{3}$

$(0,8\frac{1}{3})$

III) X-Intercepts

$y=0$

$\frac{1}{3}(x+5)^2=0$

$\frac{\frac{1}{3}(x+5)^2}{\frac{1}{3}}=\frac{0}{\frac{1}{3}}$

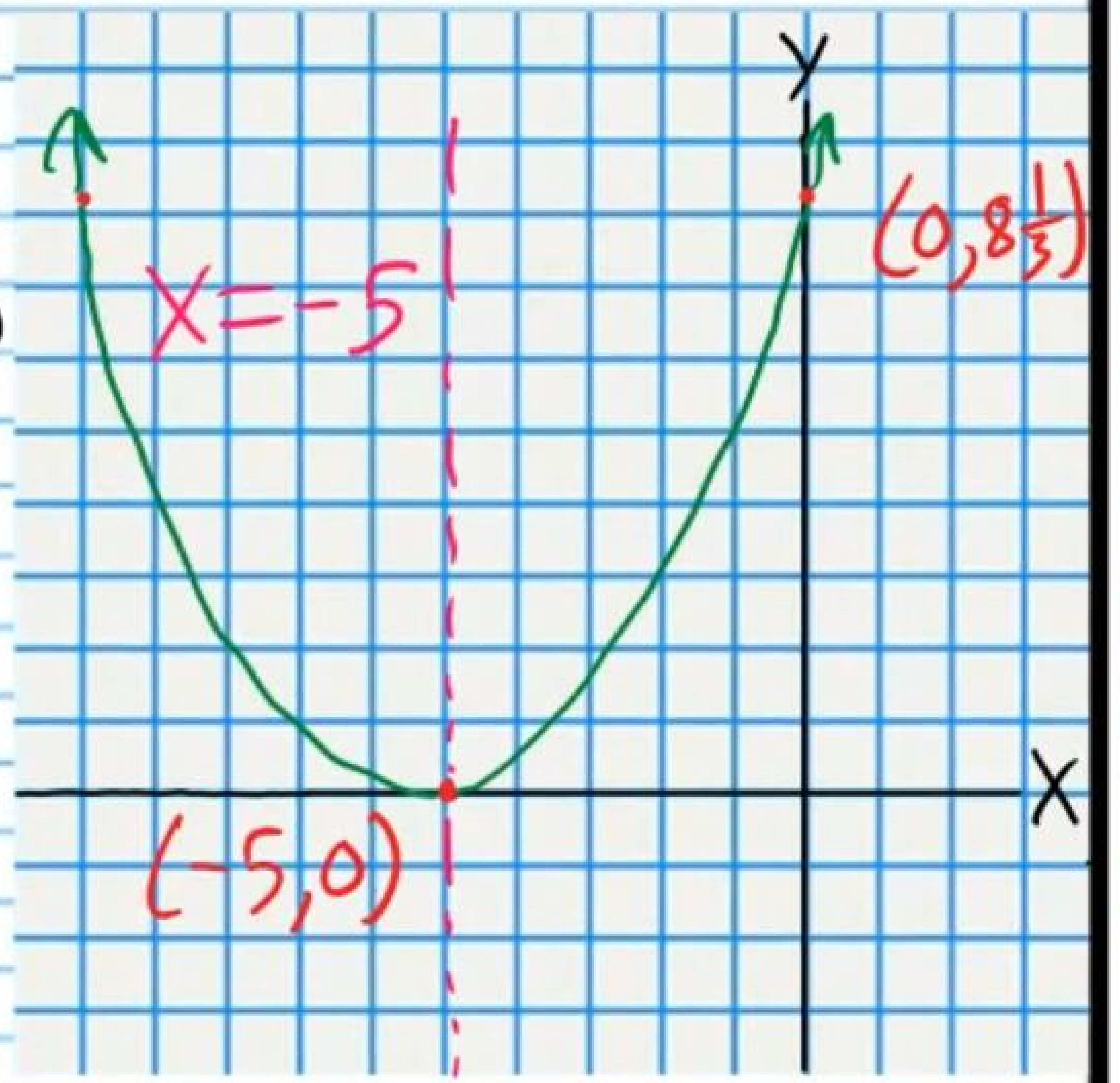
$(x+5)^2=0$

$\sqrt{(x+5)^2}=\pm\sqrt{0}$

$x+5=\pm 0$

$x+5=0$

$x=-5$ $(-5,0)$



㉑ $y=2(x-1)^2-8$

$a=2,\ h=1,\ k=-8$

I) $V_x=h=1,\ V_y=k=-8$

$V(1,-8)$

II) Y-Intercept

$x=0$

$y=2(0-1)^2-8$

$y=2(-1)^2-8$

$y=2(1)-8$

$y=2-8$

$y=-6$

$(0,-6)$

III) X-Intercepts

$y=0$

$0=2(x-1)^2-8$

$8=2(x-1)^2$

$4=(x-1)^2$

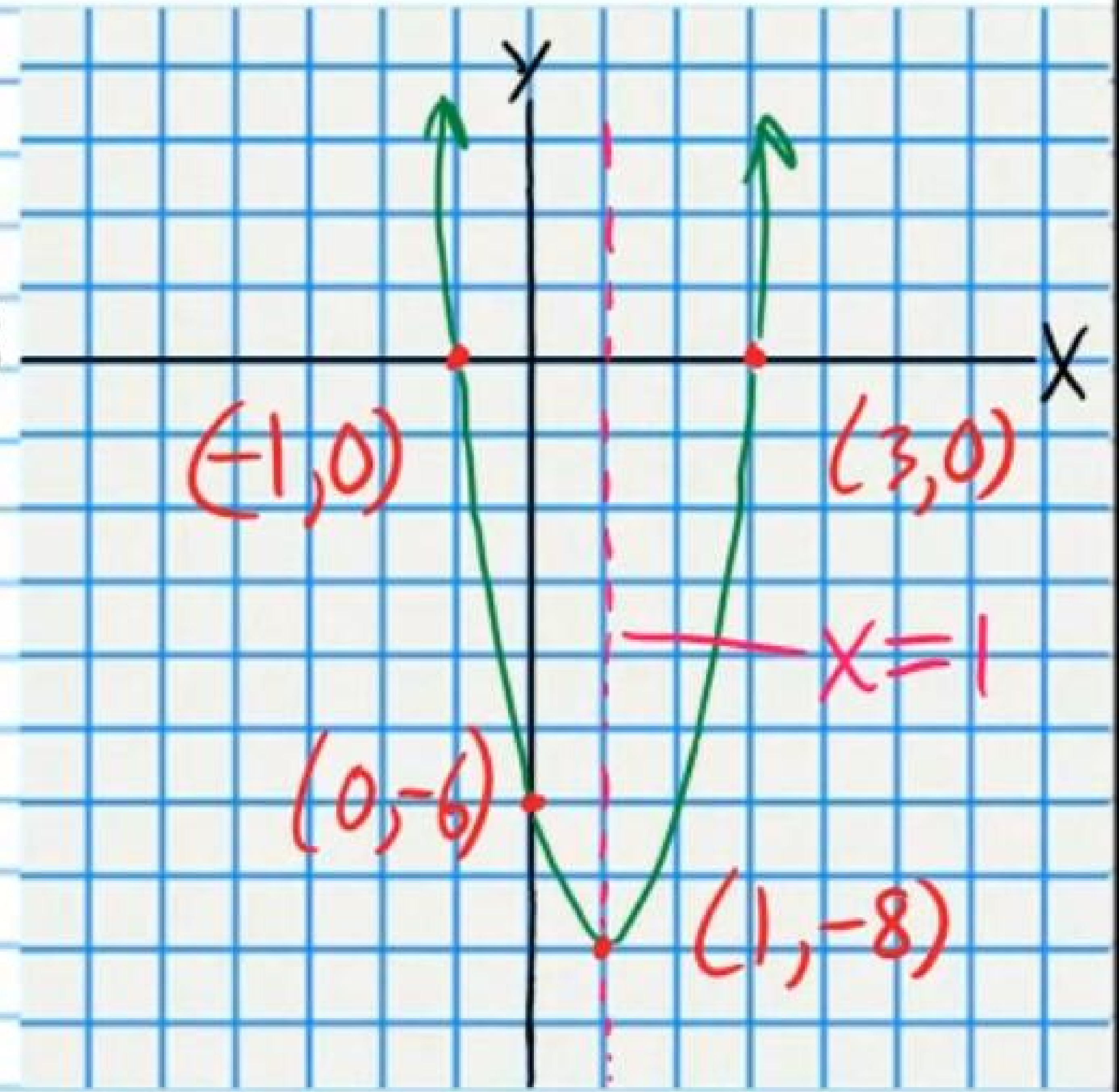
$\pm\sqrt{4}=\sqrt{(x-1)^2}$

$\pm 2=x-1$

$1\pm 2=x$

$x=3$ or $x=-1$

$(3,0)$ & $(-1,0)$

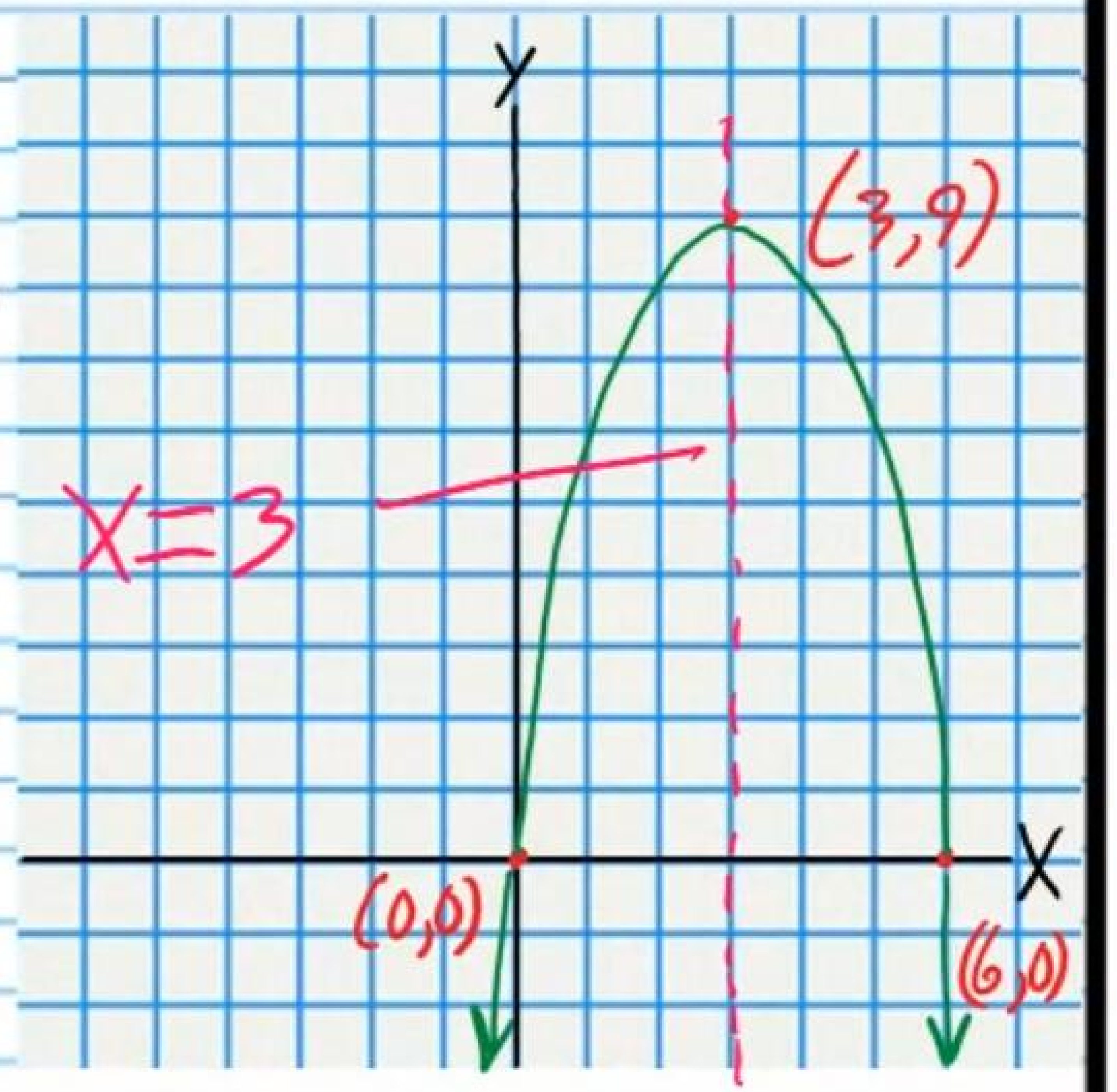


22. $y = -(x-3)^2 + 9$

$a = -1,\ h = 3,\ k = 9$

I) $V_x = h = 3,\ V_y = k = 9$

$V(3,9)$



II) Y-Intercept

$x = 0$

$y = -(0-3)^2 + 9$

$y = -(-3)^2 + 9$

$y = -(9) + 9$

$= 0$

$(0,0)$

III) X-Intercepts

$y = 0$

$0 = -(x-3)^2 + 9$

$-9 = -(x-3)^2$

$9 = (x-3)^2$

$\pm\sqrt{9} = \sqrt{(x-3)^2}$

$\pm 3 = x - 3$

$3 \pm 3 = x$

$x = 6$ or $x = 0$

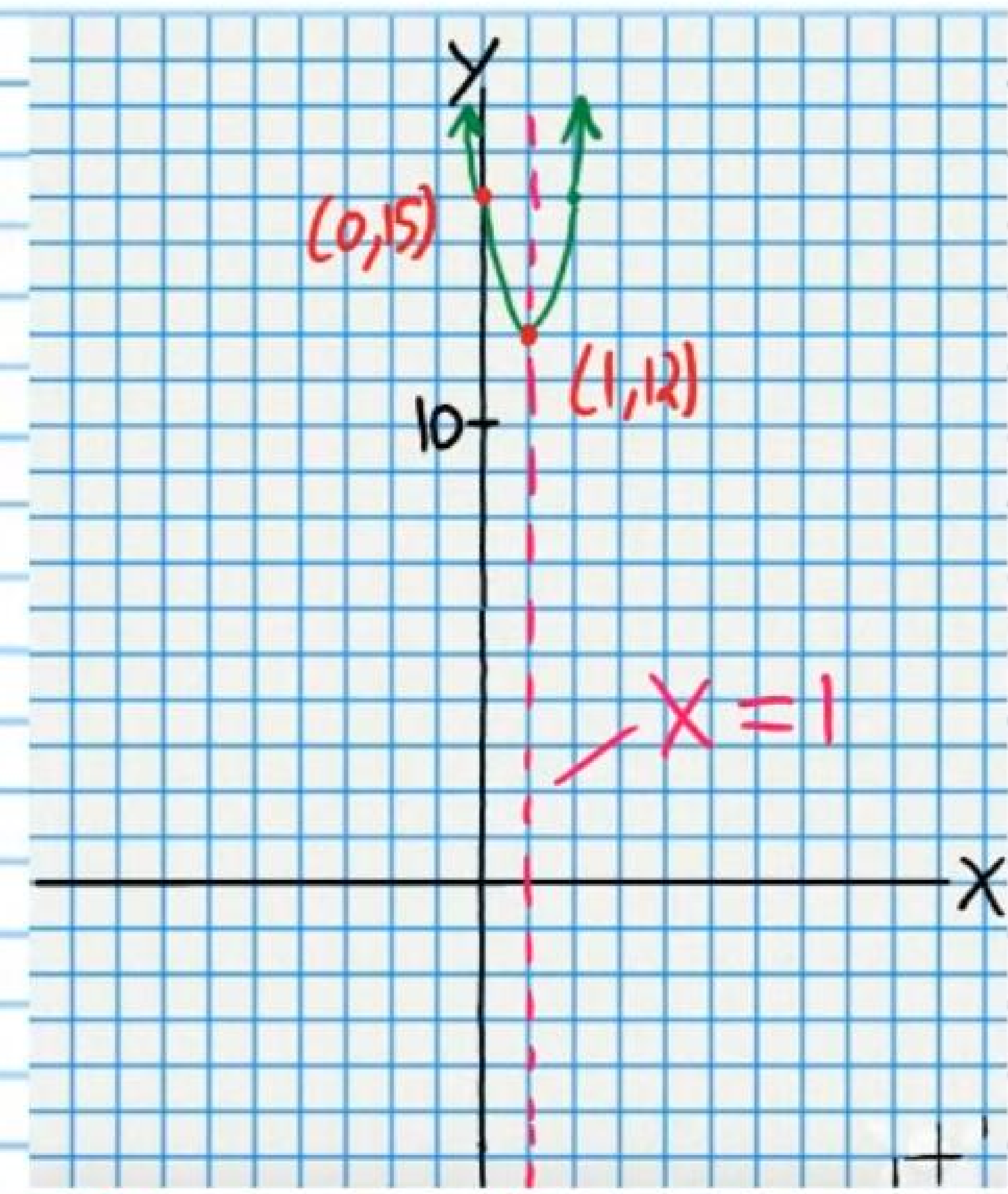
$(6,0)$ & $(0,0)$

23. $y = 3(x-1)^2 + 12$

$a = 3,\ h = 1,\ k = 12$

I) $V_x = h = 1,\ V_y = k = 12$

$V(1,12)$



No X-Intercepts

II) Y-Intercept

$x = 0$

$y = 3(0-1)^2 + 12$

$y = 3(-1)^2 + 12$

$y = 3(1) + 12$

$y = 3 + 12$

$y = 15$

$(0,15)$

III) X-Intercepts

$y = 0$

$0 = 3(x-1)^2 + 12$

$-12 = 3(x-1)^2$

$-4 = (x-1)^2$

STOP

$\pm\sqrt{-4} = \sqrt{(x-1)^2}$

$\pm i\sqrt{4} = x - 1$

$\pm i2 = x - 1$

$\pm 2i = x - 1$

$1 \pm 2i = x$

$x = 1 + 2i$ or $x = 1 - 2i$

(24) $-2(x+1)^2-18=y$

$a=-2,\ h=-1,\ k=-18$

I) $V_x=h=-1,\ V_y=k=-18$

$V(-1,-18)$

II) Y-Intercept

$x=0$

$-2(0+1)^2-18=y$

$-2(1)^2-18=y$

$-2(1)-18=y$

$-2-18=y$

$-20=y$

$(0,-20)$

III) X-Intercepts

$y=0$

$-2(x+1)^2-18=0$

$-2(x+1)^2=18$

$(x+1)^2=-9$

STOP